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Duration-Dependent Efficacy and Clinical Safety of Repeated Low-Level Red-Light Therapy for Paediatric Myopia: A Systematic Review and Meta-Analysis

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ABSTRACT

Background: Repeated low-level red-light (RLRL) therapy is a novel, non-invasive intervention for controlling paediatric myopia progression. Despite increasing clinical use, questions remain regarding the magnitude, durability, and safety of treatment effects. This study evaluated its efficacy and safety based on randomised controlled trial (RCT) evidence.

Methods: PubMed, Embase, and the Cochrane Library were searched through August 2025 for RCTs comparing RLRL with control in children (< 18 years). The primary outcome was spherical equivalent refraction (SER); secondary outcomes included axial length (AL), choroidal thickness (ChT), anterior chamber depth (ACD), central corneal thickness (CCT), lens thickness (LT), and uncorrected visual acuity (UCVA). Evidence certainty was assessed using the GRADE framework.

Results: Twenty-eight RCTs (3573 participants) were included. RLRL therapy achieved duration-dependent myopia control ($p < 0.05$), with pooled improvements versus control at 12 months of +0.68 D in SER, −0.30 mm in AL, and +26.7 μm in ChT. Subgroup analysis showed greater efficacy in children with higher baseline myopia ($p < 0.0001$), with comparable effects between 5- and 7-day regimens. UCVA was higher with RLRL therapy. ACD, CCT, and LT were unchanged. Reported adverse events were mild and transient, with no consistent structural or functional abnormalities described.

Conclusions: RLRL was associated with duration-dependent reductions in myopia progression up to 1 year. Reported structural parameters remained stable and adverse events were generally mild. Moderate-certainty evidence supports improvement in SER and ChT, whereas evidence for AL remains of lower certainty. Longer-term trials incorporating comprehensive retinal safety assessment are required.

Trial Registration: PROSPERO CRD420251112677

Abbreviations: ACD, anterior chamber depth; AL, axial length; CCT, central corneal thickness; ChT, choroidal thickness; CI, confidence interval; LT, lens thickness; MD, mean difference; RCT, randomised controlled trial; RLRL, repeated low-level red-light; SER, spherical equivalent refraction; SVS, single-vision spectacle; UCVA, uncorrected visual acuity.

Lee-Yuan Lin, Cheng-Hao Hsu, and Hung Su equally contributed to this study.

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1 | Introduction

Myopia is a refractive error in which light from distant objects is focused in front of the retina, resulting in blurred vision; this refractive error is the most common cause of vision impairment worldwide [1, 2]. Notably, the global prevalence of myopia increased from 22.9% in 2000 to 33.9% in 2020; myopia is predicted to affect 4.758 billion people by 2050, representing 49.8% of the global population, with 9.8% of the global population predicted to have high myopia [3]. Myopia typically starts in childhood and continues to progress through the teenage years [4]. High myopia is associated with an elevated risk of myopic macular degeneration, retinal detachment, cataract, glaucoma, and even irreversible visual impairment [5]. Therefore, early control of paediatric myopia may be beneficial in reducing the risk of developing high myopia and its associated complications.

To address the growing burden of paediatric myopia, several interventions for controlling its progression have been developed. These interventions include low-dose atropine eye drops, orthokeratology lenses, and myopic-defocus contact lenses or spectacles [6]. SVSs are widely used to correct short-sightedness, but they do not slow its progression [7]. Low-dose atropine (dose of 0.01%–1.0%) slows myopia progression, although its side effects, including photophobia and poor near visual acuity, may affect patient compliance [8]. Orthokeratology lenses, worn overnight to temporarily modify the curvature of the cornea by flattening it, are associated with increased risks of microbial keratitis and corneal staining and necessitate strict hygiene practices [9]. In recent years, repeated low-level red-light (RLRL) therapy has emerged as an alternative modality for paediatric myopia and has been increasingly evaluated in clinical trials, owing to its non-invasive mode of delivery and the growing body of clinical trial data evaluating its efficacy and safety [10].

RLRL therapy has been investigated for paediatric myopia control primarily over the past few years, with a rapid increase in the number of randomised controlled trials (RCTs) published recently [11, 12]. The hypothesized mechanism involves enhanced retinal blood flow and mitochondrial metabolism along with reduced scleral hypoxia and remodelling, ultimately contributing to inhibition of axial elongation [13]. RLRL therapy is generally delivered at home under parental supervision through a portable, light-emitting-diode-based device [14]. In the current RLRL protocol, which lasts several months or even years, each eye is irradiated twice daily, with each session lasting 3 min, by using low-dose red light at a wavelength of 650 ± 10 nm [15]. However, despite growing clinical adoption, important questions remain regarding the magnitude and durability of treatment effects as well as the full scope of ocular safety, particularly in light of emerging discussions surrounding retinal structural outcomes and photobiological exposure standards.

Several clinical trials have suggested that RLRL therapy slows axial elongation and paediatric myopia progression [16, 17]. Nevertheless, individual trials are often limited by modest sample sizes, different follow-up durations, and variable reporting of safety outcomes. In parallel, multiple systematic reviews and meta-analyses have been published, many of which included mixed study designs or comparatively fewer RCTs, and primarily focused on overall efficacy rather than safety or duration-dependent effects. Given the recent expansion of high-quality RCT evidence, there is a need for an updated synthesis that systematically evaluates both efficacy and safety and characterises how treatment effects evolve over time. Therefore, we conducted this systematic review and meta-analysis to evaluate the efficacy and safety of RLRL therapy in paediatric myopia, with a specific focus on duration-dependent effects, ocular safety outcomes, and evidence certainty assessed using the GRADE framework. Our aim was to update the RCT evidence base and to provide a methodologically rigorous and clinically oriented synthesis that clarifies the magnitude and duration of treatment effects, evaluates the scope of reported safety outcomes, and delineates areas where evidence remains limited.

2 | Methods

This systematic review and meta-analysis was prospectively registered at the International Prospective Register of Systematic Reviews (PROSPERO; identifier: CRD420251112677). This study was conducted in accordance with the Cochrane Handbook and in adherence to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Table S1) [18].

2.1 | Eligibility Criteria

To be included for analysis, studies had to be RCTs evaluating the efficacy of RLRL therapy in paediatric myopia control and comparing either (1) RLRL therapy alone versus observation or no active treatment, or (2) RLRL therapy combined with a baseline intervention (e.g., single-vision spectacles, atropine eye drops, or orthokeratology lenses) versus the same baseline intervention alone. Eligible trials included participants aged <18 years with myopia (≤ -0.5 D) or premyopia (> -0.5 D) and had clearly stated patient selection criterion, RLRL therapy protocol and device information, treatment adherence, interval of follow-up visits, and myopia outcome definitions. A trial was excluded if RLRL therapy was not included in the primary intervention; if the reported outcomes did not include any of the prespecified primary or secondary outcomes of interest; if the study was not an RCT; or if the publication was an abstract, letter, comment, editorial, or poster with incomplete data.

2.2 | Search Methods

Two reviewers (HS and CHH) systematically searched PubMed, Embase, and the Cochrane Library from their inception until August 2025 for relevant trials. Additional trials were identified through reference lists and by contacting experts in the field. The following terms were used in the search process: "myopia," 'nearsighted', 'low-level light therapy', and 'phototherapy'. The World Health Organization International Clinical Trials Registry Platform and the ClinicalTrials.gov registry (http://clinicaltrials.gov) were also screened. No language restrictions were applied. The full search strategies are detailed in Table S2.

2.3 | Trial Selection

The same two reviewers judged the titles and abstracts of the included trials for potential eligibility, and screened the full-text articles on the basis of the predefined eligibility criteria. Any disagreements between the reviewers were resolved through discussion with a third reviewer (LYL). To prevent data duplication, in cases in which multiple trials involved the same population or database, only one publication was selected for inclusion in our study.

2.4 | Data Extraction

The same two reviewers extracted data from each eligible trial into a standardised electronic database. Any discrepancies between the reviewers were resolved through discussion. These data included (1) study information (first author's name and publication year); (2) number and characteristics of participants (age, sex and race or ethnicity, if available); (3) inclusion and exclusion criteria; (4) intervention and control; (5) irradiation scheme of RLRL therapy; (6) length of follow-up; (7) baseline and mean changes in myopia-related outcomes such as spherical equivalent refraction (SER), axial length (AL), and choroidal thickness (ChT); and (8) other outcomes such as anterior chamber depth (ACD), central corneal thickness (CCT), lens thickness (LT), and uncorrected visual acuity (UCVA).

2.5 | Risk-of-Bias Assessment

The same two reviewers assessed the risk of bias of the included trials by using the revised tool for assessing risk of bias in randomised trials (RoB 2.0) [19]. This tool was employed to evaluate 5 domains of bias: the randomisation process, deviations from intended interventions, missing outcome data, the measurement of outcomes, and the selection of reported results. Each domain was judged as having a low risk of bias, some concerns, or a high risk of bias. Discrepancies were resolved through consultation with a third reviewer (LYL).

2.6 | Outcomes

Changes in SER (D) were used as the primary outcome to evaluate the efficacy of RLRL therapy in slowing myopia progression, with analyses conducted on the data obtained at 1, 3, 6, 9 and 12 months after treatment. The secondary outcomes were AL (mm), ChT (μm), ACD (mm), CCT (μm), LT (mm) and UCVA (logMAR).

2.7 | Certainty of Evidence Assessment

The certainty of evidence for each outcome was appraised using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach [20]. Evaluation covered five domains: risk of bias, inconsistency, indirectness, imprecision, and potential publication bias. Based on these criteria, each outcome was rated as having high, moderate, low, or very low certainty. When substantial issues were identified in any domain, the certainty level was downgraded in accordance with GRADE guidelines. The GRADE assessment was carried out independently by LYL and JWS, and disagreements were resolved through consultation with SHL. The findings were then summarised in a table generated using the GRADEpro GDT software.

2.8 | Data Synthesis and Analysis

All meta-analyses were conducted using Review Manager version 5.4 (Cochrane Collaboration, Oxford, England) and R software version 4.5.0 (R Foundation for Statistical Computing, Vienna, Austria). MDs and corresponding 95% confidence intervals (CIs) were calculated for both the primary and secondary outcomes. Statistical heterogeneity was evaluated using the I^2 statistic, which quantifies the proportion of total variability attributable to between-study differences. Significant heterogeneity was defined by an I^2 value of $>50\%$ or a Cochrane's Q test p value of <0.05. A subgroup analysis was conducted on the basis of therapy duration, treatment frequency, and participants' baseline SER. To account for duration-dependent effects, meta-regression was performed using the R package 'metafor' (version 4.8).

3 | Results

3.1 | Study Selection

Figure 1 presents a flowchart of the trial selection process. A total of 647 articles were initially identified (95 articles from PubMed, 457 articles from Embase, and 95 articles from the Cochrane Library), including 549 unique articles. Of these unique articles, 511 were excluded after their titles and abstracts were screened because they did not meet the predefined eligibility criteria. The remaining 38 articles were subjected to a full-text review, and 9 articles were excluded for the following reasons: having an incompatible study design ($n=5$) [21–25], reporting outcomes irrelevant to our study ($n=1$) [26], and utilising light wavelengths other than those of red light ($n=3$) [27–29]. Of the remaining trials, 2 involved the same study cohort [30, 31]. Ultimately, 28 RCTs involving 3573 patients were included in this meta-analysis [13–17, 30, 32–53].

3.2 | Characteristics of the Included Trials

The included trials involved between 34 and 336 participants (Table 1). The percentage of male participants varied between the trials, with the majority of trials reporting a percentage of 45%–60%. All trials except one [33] were conducted in China. The baseline population characteristics varied from premyopia to high myopia. Some trials combined RLRL therapy with atropine eye drops [17], orthokeratology lenses [40, 45], and SVSs [13, 14, 16, 30, 32–34, 36, 38, 43, 44, 46–53], whereas others applied RLRL therapy as a standalone intervention. In the majority of trials, the RLRL protocol involved red light at a wavelength of 650 ± 10 nm, with an irradiation schedule of 3 min per session, twice daily, for 5 or 7 days per week. All included trials except two explicitly reported the use of cycloplegic refraction for refractive

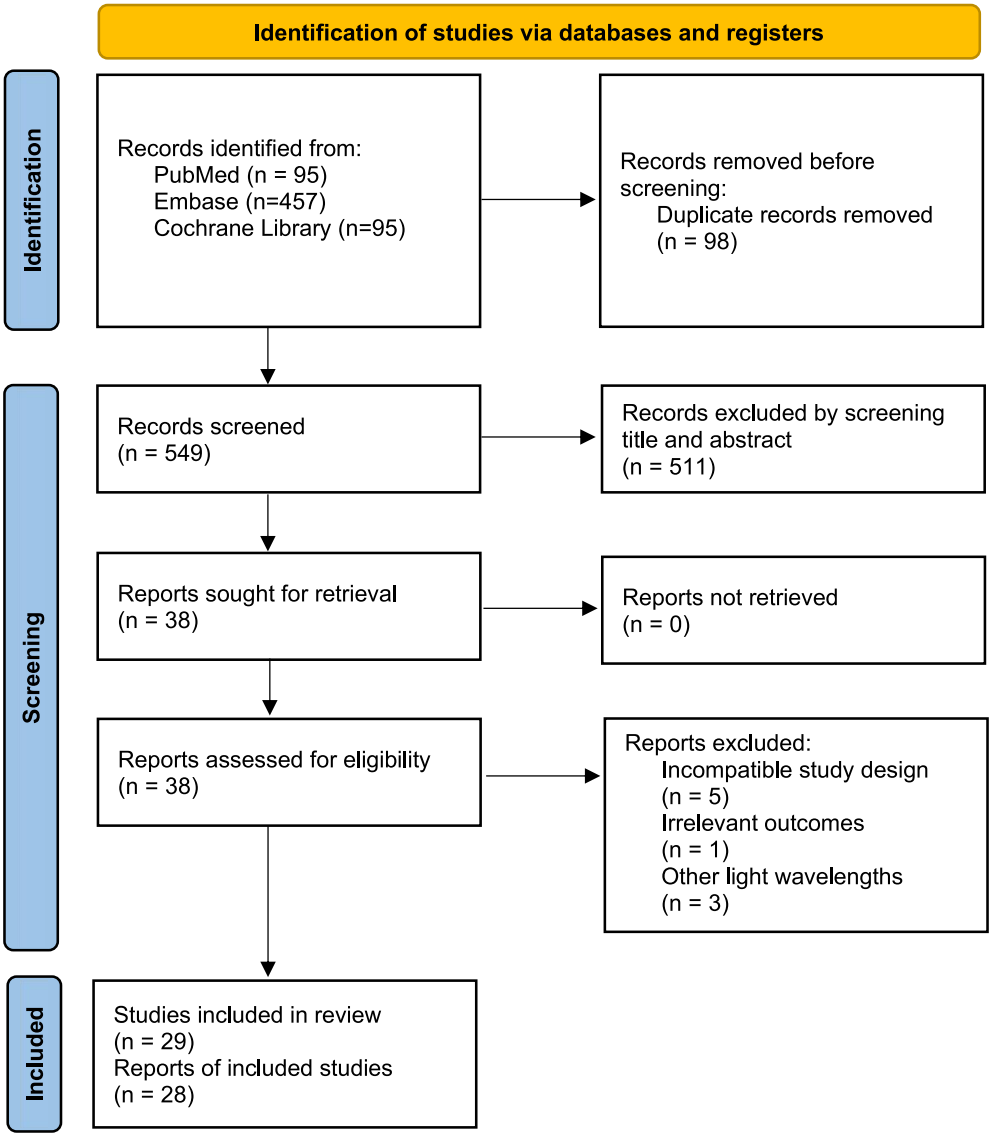


FIGURE 1 | PRISMA flowchart of the trial selection process.

assessment; one trial reported cycloplegic refraction at baseline prior to spectacle prescription but did not specify cycloplegia status during follow-up [40], whereas another did not explicitly report cycloplegic refraction at any assessment time point [14]. Follow-up visits were generally made from 1 to 12 months, with only one trial providing additional data 3 months after treatment cessation [32].

3.3 | Risk-of-Bias Assessment

Table 2 presents a summary of the risk-of-bias assessment results of the 28 included RCTs. Overall, 11 trials were judged as having some concerns, whereas the remaining 17 trials were judged as having a low risk of bias. The most common source of concern was bias due to deviations from intended interventions, primarily because these trials did not involve an intention-to-treat analysis [16, 32, 36, 39–42, 46, 49, 51, 53]. Additionally, one trial allocated participants on the basis of treatment preference rather than randomisation, thereby introducing a risk of randomisation bias [53].

3.4 | RLRL Therapy Slows Myopia Progression in a Duration-Dependent Manner

In the included RCTs, efficacy outcomes were assessed longitudinally in the same cohort of participants, allowing evaluation of how treatment effects evolved with increasing duration of RLRL exposure. Meta-analysis and meta-regression were conducted, and the results shown in Figures 2 and S1 indicated that the therapeutic effect of RLRL therapy on SER increased with the follow-up duration ($p < 0.05$, $I^2 = 97.6\%$); the mean improvements over the control were 0.36 D (95% CI = 0.29–0.44) at 6 months and 0.68 D (95% CI = 0.57–0.78) at 12 months. The therapeutic effect of RLRL on AL also strengthened over time ($p < 0.05$, $I^2 = 97.4\%$), with mean reductions of 0.19 mm (95% CI = 0.15–0.24) at 6 months and 0.30 mm (95% CI = 0.23–0.37) at 12 months relative to control. Similarly, RLRL therapy attenuated ChT thinning under a longer follow-up ($p < 0.05$, $I^2 = 86.1\%$), with MDs of 25.11 μm (95% CI = 19.42–30.81) at 6 months and 26.70 μm (95% CI = 18.64–34.77) at 12 months. Overall, RLRL therapy consistently outperformed control in slowing SER progression, reducing AL elongation, and attenuating ChT

TABLE 1 | Characteristics of the included studies.

Author (year)	Inclusion criteria	Group	*N* (%)	Age (y)	Red light wavelength (nm), power (mW)	Irradiation scheme	Follow-up visits	Baseline SER (D)	Baseline AL (mm)
Cao K et al. (2024) [13]	Aged 6–12 years, +3.00 D ≥ SER ≥ −6.00 D in both eyes, astigmatism ≤ 2.50 D, anisometropia ≤ 1.50 D	RLRL-12 mo	168 (46.4)	9.1 ± 2.0	650, not mentioned	3 min/time, 2 times/d, 7 days/week + SVS	6 and 12 mo	−1.4 ± 1.6	23.9 ± 1.1
		Control	168 (48.8)	9.0 ± 1.9	NA	SVS	6 and 12 mo	−1.3 ± 1.5	23.8 ± 1.0
Chen H et al. (2023) [32]	Aged 6–13 years, −0.75 D ≥ SER ≥ −6.00 D, astigmatism ≤ 1.50 D, anisometropia ≤ 1.50 D, mono-ocular BCVA ≥ 20/20, IOP 10–21 mmHg	RLRL-12 mo	46 (58.7)	9.00 ± 1.90	635, 0.35[b]	3 min/time, 2 times/d, 7 days/week + SVS	3, 6, 9, 12, and 15 mo	−2.54 ± 1.04	24.62 ± 0.97
		Control	40 (62.5)	8.98 ± 1.92	NA	SVS	3, 6, 9, 12, and 15 mo	−2.29 ± 0.77	24.57 ± 0.76
Chen Y et al. (2022) [16]	Aged 7–15 years, −1.00 D ≥ SER, astigmatism ≤ 2.50 D, mono-ocular BCVA ≥ 20/20	RLRL-12 mo	31 (45.2)	9.78 ± 1.58	650 ± 10, 1600 lux	3 min/time, 2 times/d, 7 days/week + SVS	1, 3, 6, and 12 mo	−2.60 ± 1.17	24.48 ± 0.79
		0.01% atropine	31 (54.8)	10.31 ± 1.90	NA	0.01% atropine once every night + SVS	1, 3, 6, and 12 mo	−2.59 ± 1.24	24.67 ± 0.98
Deen N et al. (2025) [33]	Aged 8–13 years, SER −0.50 to −5.00 D, astigmatism ≤ 2.50 D, anisometropia ≤ 1.50 D, BCVA ≥ 20/20	RLRL-3 mo	16 (56.25)	11.5 + 1.55	650 ± 10, 2[b]	3 min/time, 2 times/d, 5 days/week + SVS	1, 3, 6, and 12 mo	−2.18 + 0.86	24.2 + 0.69
		Control	18 (55.56)	11.9 + 1.39	NA	SVS	1, 3, 6, and 12 mo	−2.24 + 1.06	24.6 + 0.75
Dong J et al. (2023) [34]	Aged 7–12 years, −0.50 D ≥ SER, astigmatism ≤ 1.50 D, anisometropia ≤ 1.50 D	RLRL-6 mo	56 (46.4)	10.3 ± 2.07	650, 0.29[a]	3 min/time, 2 times/d, 7 days/week + SVS	6 mo	−3.13 ± 1.91	24.7 ± 1.04
		10% RLRL	55 (53.6)	9.86 ± 1.41	650, 0.03[a]	3 min/time, 2 times/d, 7 days/week + SVS	6 mo	−2.82 ± 1.86	24.6 ± 0.96
He X et al. (2023) [35]	Aged 6–11 years, +0.50 D ≥ SER ≥ −0.50 D in the more myopic eye, at least 1 parent with SER ≤ −3.00 D, astigmatism ≤ 1.50 D, anisometropia ≤ 1.50 D	RLRL-12 mo	139 (51.1)	8.28 ± 1.10	650 ± 10, not mentioned	3 min/time, 2 times/d, 5 days/week	3, 6, 9, and 12 mo	0.14 ± 0.30	23.36 ± 0.68
		Control	139 (48.9)	8.31 ± 1.07	NA	NA	3, 6, 9, and 12 mo	0.16 ± 0.28	23.30 ± 0.69
Jiang Y et al. (2022) [30]	Aged 8–13 years, −1.00 D ≥ SER ≥ −5.00 D, astigmatism ≤ 2.50 D, anisometropia ≤ 1.50 D, mono-ocular BCVA ≥ 20/20	RLRL-12 mo	119 (47.9)	10.4 (8.0, 13.0)	650 ± 10, 0.29[a] (1600 lux)	3 min/time, 2 times/d, 5 days/week + SVS	1, 3, 6, and 12 mo	−2.49 ± 0.92	24.54 ± 0.67
		Control	145 (50.3)	10.5 (8.1, 13.0)	NA	SVS	1, 3, 6, and 12 mo	−2.67 ± 1.06	24.62 ± 0.86

(Continues)

TABLE 1 | (Continued)

Author (year)	Inclusion criteria	Group	*N* (%)	Age (y)	Red light wavelength (nm), power (mW)	Irradiation scheme	Follow-up visits	Baseline SER (D)	Baseline AL (mm)
Jiang Z et al. (2025) [14]	Aged 8–15 years, −1.00 D ≥ SER ≥ −5.00 D, astigmatism < 2.50 D, anisometropia ≤ 1.50 D, mono-ocular BCVA ≥ 20/20	RLRL-9 mo	35 (54.28)	11.32 ± 2.05	650 ± 10, 1600 lux	3 min/time, 2 times/d, 5 days/week + SVS	1, 3, 6, 9, 10, 11, and 12 mo	−2.37 ± 0.69	24.59 ± 0.77
		Control	35 (54.28)	11.37 ± 2.08	NA	SVS	1, 3, 6, 9, 10, 11, and 12 mo	−2.21 ± 0.71	24.58 ± 0.82
Li L et al. (2025) [37]	Aged 4–8 years, with SER ≤ −0.50 D after cycloplegia, clinical diagnosis of myopic amblyopia and without pathological myopia, astigmatism ≤ 2.50 D	RLRL-6 mo	31 (29)	5.52 ± 1.061	650 ± 10, 2[b] (1600 lux)	3 min/time, 2 times/d, 7 days/week	1, 3, and 6 mo	−3.75 (−5.75, −2.75)	24.11 ± 0.95
		Control	35 (51)	5.00 ± 1.213	NA	SVS	1, 3, and 6 mo	−2.75 (−4.50, −1.75)	24.01 ± 0.70
Liu G et al. (2024) [39]	Aged 8–13 years, −1.00 D ≥ SER ≥ −6.00 D, astigmatism < 1.50 D, anisometropia ≤ 1.50 D, mono-ocular BCVA ≥ 20/20	Myopia-RLRL	32 (50)	9.37 ± 1.69	650 ± 10, 1600 lux	3 min/time, 2 times/d, 7 days/week	1, 3, 6, 9, and 12 mo	−2.91 ± 1.27	24.71 ± 0.92
		Myopia-control	36 (47.22)	9.55 ± 1.13	NA	NA	1, 3, 6, 9, and 12 mo	−2.61 ± 0.98	24.58 ± 0.64
	Aged 8–13 years, +0.75 D ≥ SER ≥ −0.50 D, astigmatism < 1.50 D, anisometropia ≤ 1.50 D, mono-ocular BCVA ≥ 20/20	Premyopia-RLRL	40 (55)	8.95 ± 0.87	650 ± 10, 1600 lux	3 min/time, 2 times/d, 7 days/week	1, 3, 6, 9, and 12 mo	0.36 ± 0.32	23.40 ± 0.63
		Premyopia-control	36 (47.22)	8.94 ± 1.09	NA	NA	1, 3, 6, 9, and 12 mo	0.37 ± 0.30	23.30 ± 0.78
Liu G et al. (2025) [38]	Aged 7–12 years, −6.00 D ≥ SER, astigmatism < 2.50 D, anisometropia ≤ 1.50 D, mono-ocular BCVA ≥ 20/20	RLRL-12 mo	130 (46.92)	10.1 ± 1.7	650 ± 10, 1600 lux	3 min/time, 2 times/d, 7 days/week + SVS	1, 3, 6, 9, and 12 mo	−7.75 ± 1.91	26.50 ± 1.03
		Control	68 (44.12)	10.5 ± 1.6	NA	SVS	1, 3, 6, 9, and 12 mo	−7.65 ± 1.70	26.38 ± 0.99
Liu Y et al. (2024) [40]	Aged 8–16 years, SER −1.00 to −6.00 D, astigmatism < 0.75 D, anisometropia < 1.00 D, BCVA > 1.0 in both eyes	RLRL-12 mo	57 (45.61)	11.57 ± 2.39	650, 2[b]	RLRL (3 min/time, 2 times/d, 5 days/week) + orthokeratology lenses (8–10 h)	1, 3, 6, and 12 mo	−2.67 ± 0.92	23.76 ± 0.42
		Control	49 (46.94)	11.09 ± 2.73	NA	Orthokeratology lenses (8–10 h)	1, 3, 6, and 12 mo	−2.63 ± 0.89	23.73 ± 0.39
Liu Z et al. (2024) [41]	Aged 7–12 years, premyopia (−0.50 D < SER < +0.75 D), astigmatism < 2.00 D	RLRL-12 mo	47 (55.81)	8.98 ± 1.31	650 ± 10, 0.29[a]	RLRL (3 min/time, 2 times/d, 7 days/week)	1, 3, 6, 9, and 12 mo	0.17 ± 0.35	23.57 ± 0.78
		Control	47 (52.38)	8.95 ± 1.52	NA	NA	1, 3, 6, 9, and 12 mo	0.30 ± 0.35	23.30 ± 0.73

(Continues)

TABLE 1 | (Continued)

Author (year)	Inclusion criteria	Group	*N* (%)	Age (y)	Red light wavelength (nm), power (mW)	Irradiation scheme	Follow-up visits	Baseline SER (D)	Baseline AL (mm)
Pang X et al. (2025) [17]	Aged 6–12 years, −0.75 D ≥ SER ≥ −5.00 D in both eyes, astigmatism < 2.00 D, anisometropia ≤ 2.00 D, mono-ocular BCVA ≥ 20/20, IOP < 21 mmHg	RLRL-atropine group	45 (46.7)	9.27 ± 1.45	650 ± 10, 1600 lux, 0.01% atropine eye drops	3 min/time, 2 times/d, 7 days/week → SVS → once every night	6, 7 and 13 mo	−2.43 ± 1.13	24.33 ± 0.81
		Atropine-RLRL group	42 (54.8)	8.98 ± 1.27	0.01% atropine eye drops, 650 ± 10	Once every night → SVS → 3 min, 2 times, 7 days/week	6, 7 and 13 mo	−2.28 ± 1.04	24.52 ± 0.82
Shang L et al. (2024) [42]	Aged 6–12 years, +0.50 D ≥ SER > −0.75 D, astigmatism < 1.00 D, anisometropia ≤ 1.50 D, mono-ocular BCVA ≥ 20/20	RLRL-6 mo	32 (46.88)	8.76 ± 1.24	650 ± 10, 0.85[b]	3 min/time, 2 times/d, 7 days/week	6 and 7 mo	−0.24 ± 0.29	23.62 ± 0.77
		0.01% atropine	30 (53.33)	8.79 ± 1.14	NA	0.01% atropine once every night	6 and 7 mo	−0.22 ± 0.30	23.59 ± 0.74
Tian L et al. (2022) [36]	Aged 6–12 years, −0.50 D ≥ SER ≥ −6.00 D in both eyes, astigmatism ≤ 2.50 D, anisometropia ≤ 1.50 D	RLRL-6 mo	112 (49.11)	9.66 ± 1.65	650, not mentioned	3 min/time, 2 times/d, 7 days/week + SVS	6 mo	−2 (−3.25, −1.25)	24.31 ± 0.92
		Control	112 (50.89)	9.47 ± 1.59	NA	SVS	6 mo	−2 (−2.75, −1.25)	24.20 ± 0.85
Tian L et al. (2023) [43]	Aged 6–12 years, +3.00 D ≥ SER ≥ −0.50 D in both eyes, astigmatism ≤ 2.50 D, anisometropia ≤ 1.50 D	RLRL-6 mo	56 (41.07)	7.7 ± 1.1	650, not mentioned	3 min/time, 2 times/d, 7 days/week + SVS	6 mo	0.25 (−0.25, 0.75)	23.1 ± 0.8
		Control	56 (44.64)	7.9 ± 1.8	NA	SVS	6 mo	0.25 (0.00, 0.75)	23.1 ± 0.7
Wei H et al. (2025) [44]	Aged 6–14 years, SER −6.00 to +0.50 D, astigmatism ≤ 1.50 D, anisometropia ≤ 1.00 D, BCVA ≥ 0.8	RLRL-12 mo	100 (47)	9.56 ± 1.97	650 ± 10, 2[b] (1600 lx)	3 min/time, 2 times/d, 5 days/week + SVS; RLRL	1, 3, 6 and 12 mo	−0.94 ± 1.62	23.72 ± 0.89
		Control	100 (51)	8.97 ± 1.90	NA	SVS; none	1, 3, 6 and 12 mo	−0.65 ± 1.56	23.39 ± 0.99
Xiang K et al. (2025) [15]	Aged 6–10 years, +0.50 D ≥ SER > −0.50 D in at least one eye, astigmatism ≤ 1.50 D, anisometropia ≤ 1.50 D	RLRL-12 mo	55 eyes (47.27)	8.27 ± 1.10	Not mentioned	3 min/time, 2 times/d, 5 days/week	3, 6, 9 and 12 mo	0.20 ± 0.28	23.30 ± 0.56
		Control	232 eyes (48.71)	8.32 ± 1.09	NA	NA	3, 6, 9 and 12 mo	0.21 ± 0.27	23.29 ± 0.67
Xiong R et al. (2023) [46]	Aged 8–13 years, −1.00 D ≥ SER ≥ −5.00 D, astigmatism ≤ 2.50 D, anisometropia ≤ 1.50 D, mono-ocular BCVA ≥ 20/20	RLRL-12 mo	60 (46.67)	10.52 ± 1.53	650, not mentioned	3 min/time, 2 times/d, 5 days/week + SVS	1, 3, 6, and 12 mo	−2.30 ± 0.85	24.52 ± 0.68
		Control	60 (45)	10.37 ± 1.61	NA	SVS	1, 3, 6 and 12 mo	−2.58 ± 1.15	24.66 ± 0.89

(Continues)

TABLE 1 | (Continued)

Author (year)	Inclusion criteria	Group	*N* (%)	Age (y)	Red light wavelength (nm), power (mW)	Irradiation scheme	Follow-up visits	Baseline SER (D)	Baseline AL (mm)
Xiong R et al. (2024) [45]	Aged 8–13 years, wearing orthokeratology lenses for 1 year, −1.00 D ≥ SER ≥ −5.00 D, astigmatism ≤ 1.50 D, anisometropia ≤ 1.50 D, mono-ocular BCVA ≥ 20/20	RLRL-12 mo	30 (40)	9.85 ± 0.95	650 ± 10, 0.29[a]	3 min/time, 2 times/d, 7 days/week + orthokeratology lens 8 h/night	1, 3, 6 and 12 mo	−2.15 ± 0.84	25.03 ± 0.80
		Control	17 (52.9)	10.13 ± 1.02	NA	Orthokeratology lens 8 h/night	1, 3, 6 and 12 mo	−2.29 ± 0.92	24.97 ± 1.09
Xiong Y et al. (2024) [47]	Aged 6–14 years, −0.50 D ≥ SER ≥ −6.00 D, astigmatism ≤ 3.00 D, anisometropia < 2.50 D, mono-ocular BCVA ≥ 20/32	RLRL-6 mo	36 (47.22)	8.83 ± 2.06	650, 0.178[a] (700 lx)	3 min/time, 2 times/d, 7 days/week + SVS	1, 3 and 6 mo	−2.47 ± 1.39	24.38 ± 0.87
		Control	37 (59.46)	9.00 ± 2.00	NA	SVS	1, 3 and 6 mo	−2.22 ± 0.72	24.47 ± 0.58
Xu Y et al. (2024) [48]	Aged 6–16 years, SER ≥ −4.00 D in at least one eye, astigmatism ≤ 2.00 D, anisometropia ≤ 3.00 D, BCVA ≥ 20/32	RLRL-12 mo	97 (65)	10.40 ± 2.4	650 ± 10, 0.29[a]	3 min/time, 2 times/d, 7 days/week + SVS	1, 3, 6, 9 and 12 mo	−5.88 ± 1.69	25.93 ± 1.03
		Control	95 (46.3)	11.20 ± 2.1	NA	SVS	1, 3, 6, 9 and 12 mo	−5.75 ± 1.17	25.72 ± 0.87
Yang K et al. (2025) [49]	Aged 8–10 years, SER −1 to −1.5 D, astigmatism ≤ 1.25 D	RLRL-12 mo	26 (61.5)	9.00 (8.00–10.00)	650 ± 10, 0.39[b]	3 min/time, 2 times/d, 7 days/week + SVS	12 mo	−1.13 (−1.38 to −1.00)	24.30 ± 0.87
		Control	26 (50)	9.00 (8.00–10.00)	NA	SVS	12 mo	−1.13 (−1.25 to −1.00)	23.93 ± 0.66
Yang Y et al. (2025) [50]	Aged 6–14 years, SER −1.00 to −6.00 D, astigmatism < 2.00 D, IOP 10 to 21, CCT > 0.45 mm	RLRL-6 mo	30 (43.33)	9.98 ± 3.41	650, 2[b]	3 min/time, 2 times/d, 5 days/week + SVS	3 and 6 mo	−4.91 ± 1.06	NA
		Control	30 (40)	10.17 ± 3.26	NA	SVS	3 and 6 mo	−4.83 ± 1.12	NA
Zhou L et al. (2023) [51]	Aged 3–16 years, −0.50 D ≥ SER ≥ −8.00 D in either eye, astigmatism ≤ 2.50 D, mono-ocular BCVA ≥ 20/20	RLRL-12 mo	24 (62.5)	10.17 ± 1.55	650 ± 10, 0.35[b] (400 lx)	3 min/time, 2 times/d, 7 days/week + SVS	3, 6, 9 and 12 mo	−2.93 ± 1.87	24.58 ± 1.16
		Control	19 (52.63)	9.00 ± 1.20	NA	SVS		−2.11 ± 1.21	24.38 ± 0.87
Zhou W et al. (2024) [52]	Aged 6–15 years, SER ≤ −0.50 D in both eyes, astigmatism < 2.50 D, IOP 10 to 21 mmHg, BCVA ≥ 20/25 in both eyes	RLRL-6 mo (0.37 ± 0.02 mW)	50 (60)	8.58 ± 1.46	650, 0.37[b]	RLRL (3 min, 2 times, 7 days/week) + SVS	1, 3 and 6 mo	−1.72 ± 0.91	24.19 ± 0.79
		RLRL-6 mo (0.60 ± 0.2 mW)	50 (50)	8.82 ± 1.45	650, 0.60[b]	RLRL (3 min, 2 times, 7 days/week) + SVS	1, 3 and 6 mo	−2.01 ± 0.87	24.12 ± 0.88
		RLRL-6 mo (1.20 mW)	50 (52)	8.78 ± 1.49	650, 1.20[b]	RLRL (3 min, 2 times, 7 days/week) + SVS	1, 3 and 6 mo	−2.08 ± 1.33	24.45 ± 0.91
		Control	50 (58)	9.03 ± 1.63	NA	SVS	1, 3 and 6 mo	−2.10 ± 0.90	24.41 ± 0.90

(Continues)

TABLE 1 | (Continued)

Author (year)	Inclusion criteria	Group	*N* (%)	Age (y)	Red light wavelength (nm), power (mW)	Irradiation scheme	Follow-up visits	Baseline SER (D)	Baseline AL (mm)
Zhu M et al. (2024) [53]	Aged 6–14 years, −0.50 D ≥ SER ≥ −6.00 D, monocular BCVA ≥ 20/20	RLRL-12 mo	53 (56.6)	8.96 ± 2.19	650 ± 10, 2[b]	3 min/time, 2 times/d, 5 days/week + SVS	3, 6 and 12 mo	−3.02 ± 1.80	24.66 ± 0.93
		Control	55 (56.36)	8.47 ± 2.10	NA	SVS	3, 6 and 12 mo	−2.85 ± 1.71	24.40 ± 1.02

Abbreviations: AL, axial length; BCVA, best-corrected visual acuity; CCT, central corneal thickness; D, diopter; IOP, intraocular pressure; mo, month(s); NA, not applicable; RLRL, repeated low-level red-light therapy; SER, spherical equivalent refraction; SVS, single-vision spectacle(s); y, year(s).
[a]Power of light entering the pupil.
[b]Power of light emitted from the device.

thinning, with progressively greater effects observed with longer follow-up within the same cohorts.

3.5 | RLRL Therapy Is More Effective in Children With Higher Baseline Myopia

To explore potential modifiers of treatment efficacy, subgroup analyses were conducted on the basis of baseline SER levels (Figure 3A). Patients were subdivided into the following categories: premyopia (greater than −0.5 D), moderate myopia (−0.5 to −5.0 D), and high myopia (less than −5.0 D) [54, 55]. At 6 months, children with premyopia had a pooled MD of 0.23 D (95% CI = 0.16–0.31, $I^2 = 57\%$), children with moderate myopia had an MD of 0.36 D (95% CI = 0.25–0.47, $I^2 = 96\%$), and children with high myopia had the highest MD of 0.66 D (95% CI = 0.56–0.75, $I^2 = 73\%$). At 12 months, the corresponding MDs were 0.42 D (95% CI = 0.24–0.60, $I^2 = 72\%$) for premyopia, 0.69 D (95% CI = 0.51–0.87, $I^2 = 97\%$) for moderate myopia, and 0.90 D (95% CI = 0.78–1.01, $I^2 = 84\%$) for high myopia. Subgroup analysis at both time points confirmed that baseline SER significantly influenced the treatment response ($p < 0.00001$ at 6 months and $p < 0.0001$ at 12 months), with the benefits being greater in children with higher myopia. The AL and ChT outcomes had similar trends to the SER outcomes, with greater treatment efficacy observed in children with higher baseline myopia (Figure S2). Notably, treatment efficacy increased over time in all baseline SER groups, with the most pronounced benefits discovered in patients with high myopia, confirming the duration-dependent effect of RLRL therapy.

3.6 | Subgroup Analysis Confirms Comparable Myopia Control of 5- and 7-Day RLRL Regimens

A subgroup analysis was conducted to compare the effects of RLRL therapy administered 5 days versus 7 days per week, with treatment efficacy assessed through SER change (Figure 3B). At 6 months, the pooled MD for SER improvement was found to be 0.34 D (95% CI = 0.14–0.54, $I^2 = 98\%$) for the 5-day group and 0.37 D (95% CI = 0.27–0.47, $I^2 = 96\%$) for the 7-day group. At 12 months, the corresponding pooled MDs were 0.60 D (95% CI = 0.28–0.91, $I^2 = 98\%$) and 0.70 D (95% CI = 0.57–0.84, $I^2 = 95\%$), respectively. Subgroup analysis revealed no significant difference between dosing frequencies for either time point ($p = 0.76$ at 6 months and $p = 0.55$ at 12 months), indicating that the 2 regimens had comparable efficacy. Consistent with the SER findings, comparable efficacy of the two regimens was found for the AL and ChT outcomes (Figure S3). Similar to the aforementioned findings, the treatment effects strengthened over time in both groups, demonstrating the consistent duration-dependent benefit of RLRL therapy regardless of dosing frequency.

3.7 | Reported Adverse Events, Anterior Segment Parameters, and Visual Outcomes

Consistent with prior safety evidence [56], across the analysed RCTs, no severe treatment-related adverse events were reported (Table 3). However, several studies reported treatment

TABLE 2 | Quality assessment of the included studies.

Author (year)	Risk of bias arising from the randomisation process	Risk of bias due to deviations from the intended interventions	Risk of bias due to missing outcome data	Risk of bias in measurement of the outcome	Risk of bias in selection of the reported result	Overall risk of bias
Cao K et al. (2024) [13]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Chen H et al. (2023) [32]	Low risk	Some concerns	Low risk	Low risk	Low risk	Some concerns
Chen Y et al. (2022) [16]	Low risk	Some concerns	Low risk	Low risk	Low risk	Some concerns
Deen N et al. (2025) [33]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Dong J et al. (2023) [34]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
He X et al. (2023) [35]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Jiang Y et al. (2022) [30]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Jiang Z et al. (2025) [14]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Li L et al. (2025) [37]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Liu G et al. (2024) [39]	Low risk	Some concerns	Low risk	Low risk	Low risk	Some concerns
Liu G et al. (2025) [38]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Liu Y et al. (2024) [40]	Low risk	Some concerns	Low risk	Low risk	Low risk	Some concerns
Liu Z et al. (2024) [41]	Low risk	Some concerns	Low risk	Low risk	Low risk	Some concerns
Pang X et al. (2025) [17]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Shang L et al. (2024) [42]	Low risk	Some concerns	Low risk	Low risk	Low risk	Some concerns
Tian L et al. (2022) [36]	Low risk	Some concerns	Low risk	Low risk	Low risk	Some concerns
Tian L et al. (2023) [43]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Wei H et al. (2025) [44]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Xiang K et al. (2025) [15]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Xiong R et al. (2023) [46]	Low risk	Some concerns	Low risk	Low risk	Low risk	Some concerns
Xiong R et al. (2024) [45]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Xiong Y et al. (2024) [47]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Xu Y et al. (2024) [48]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Yang K et al. (2025) [49]	Low risk	Some concerns	Low risk	Low risk	Low risk	Some concerns
Yang Y et al. (2025) [50]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk

(Continues)

TABLE 2 | (Continued)

Author (year)	Risk of bias arising from the randomisation process	Risk of bias due to deviations from the intended interventions	Risk of bias due to missing outcome data	Risk of bias in measurement of the outcome	Risk of bias in selection of the reported result	Overall risk of bias
Zhou L et al. (2023) [51]	Low risk	Some concerns	Low risk	Low risk	Low risk	Some concerns
Zhou W et al. (2024) [52]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Zhu M et al. (2024) [53]	Some concerns	Some concerns	Low risk	Low risk	Low risk	Some concerns

discontinuations, including due to afterimages persisting beyond 6 min [35, 53] and due to uncomfortable brightness [13, 30]. Other complaints were infrequent and generally mild.

In this study, the pooled MDs for ACD were found to be −0.01 mm (95% CI = −0.03–0.00) at 6 months and −0.01 mm (95% CI = −0.02–0.00) at 12 months, representing a borderline and clinically minimal difference ($p = 0.08$) between RLRL therapy and control (Figure 4A). No statistically significant differences were observed for CCT, with the pooled MDs for CCT of 0.47 μm (95% CI = −0.18–1.11) at 6 months and 2.30 μm (95% CI = −0.37–4.98) at 12 months (Figure 4B). Similarly, LT remained unchanged, with pooled MDs of 0.00 mm (95% CI = −0.02–0.02) at 6 months and −0.00 mm (95% CI = −0.02 to 0.01) at 12 months (Figure 4C).

With respect to visual function, RLRL therapy was associated with improved UCVA at 12 months, with pooled risk ratios of 2.33 for greater than 2-line improvement (95% CI = 1.59–3.40) and 0.58 for greater than 2-line worsening (95% CI = 0.40–0.84) compared to control groups (Figure 4D).

Overall, based on trial-reported data, RLRL therapy was associated with improved visual outcomes and stable anterior segment biometric parameters within the follow-up periods studied.

3.8 | Sensitivity Analysis and Publication Bias Assessment

A leave-one-out sensitivity analysis was conducted for the 12-month SER, AL, and ChT outcomes (Table S3) to evaluate the robustness of the pooled results. Omission of individual studies did not substantially influence the overall effect estimate, supporting the robustness of the findings. Additionally, Egger's test for 12-month SER and ChT yielded p values greater than 0.05, indicating no significant evidence of publication bias. In contrast, Egger's test for 12-month AL produced a p value less than 0.05, suggesting the presence of potential publication bias among the included studies. Funnel plots are provided in Figure S4.

3.9 | Certainty of Evidence

The overall certainty of evidence was assessed using the GRADE framework (Table 4). For SER at 12 months, the certainty of evidence was moderate, mainly due to inconsistency ($I^2 > 90\%$). For AL at 12 months, the certainty of evidence was low, attributable to publication bias and inconsistency. For ChT at 12 months, the certainty of evidence was moderate, with downgrading due to inconsistency.

4 | Discussion

This systematic review and meta-analysis of 28 RCTs revealed that RLRL therapy provided duration-dependent benefits in myopia control across all follow-up periods. Specifically, RLRL therapy significantly slowed myopia progression, inhibited axial

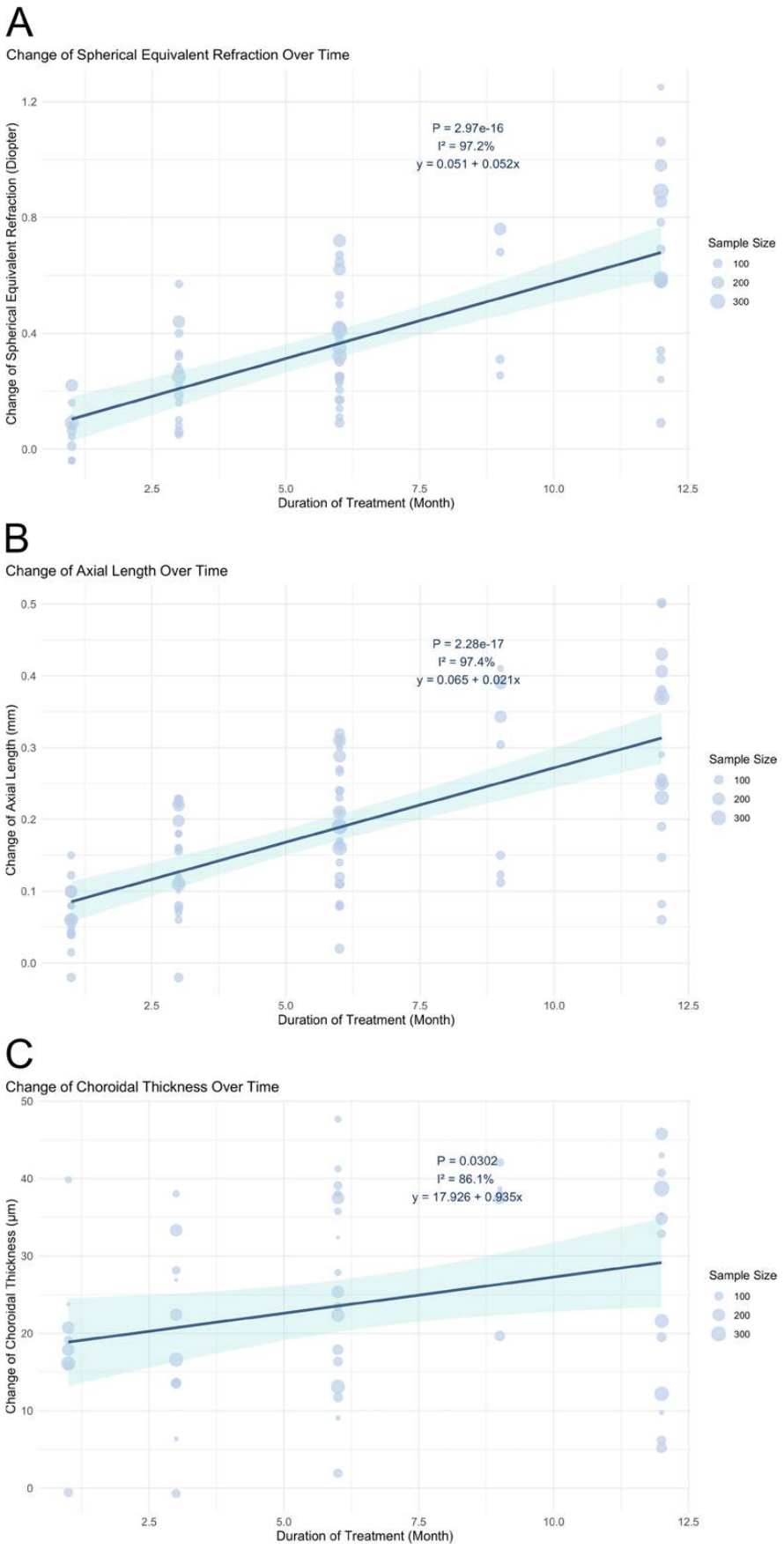


FIGURE 2 | Meta-regression analysis of the association between treatment duration and ocular parameters. (A) Change in SER versus treatment duration. (B) Change in AL versus treatment duration. (C) Change in ChT versus treatment duration.

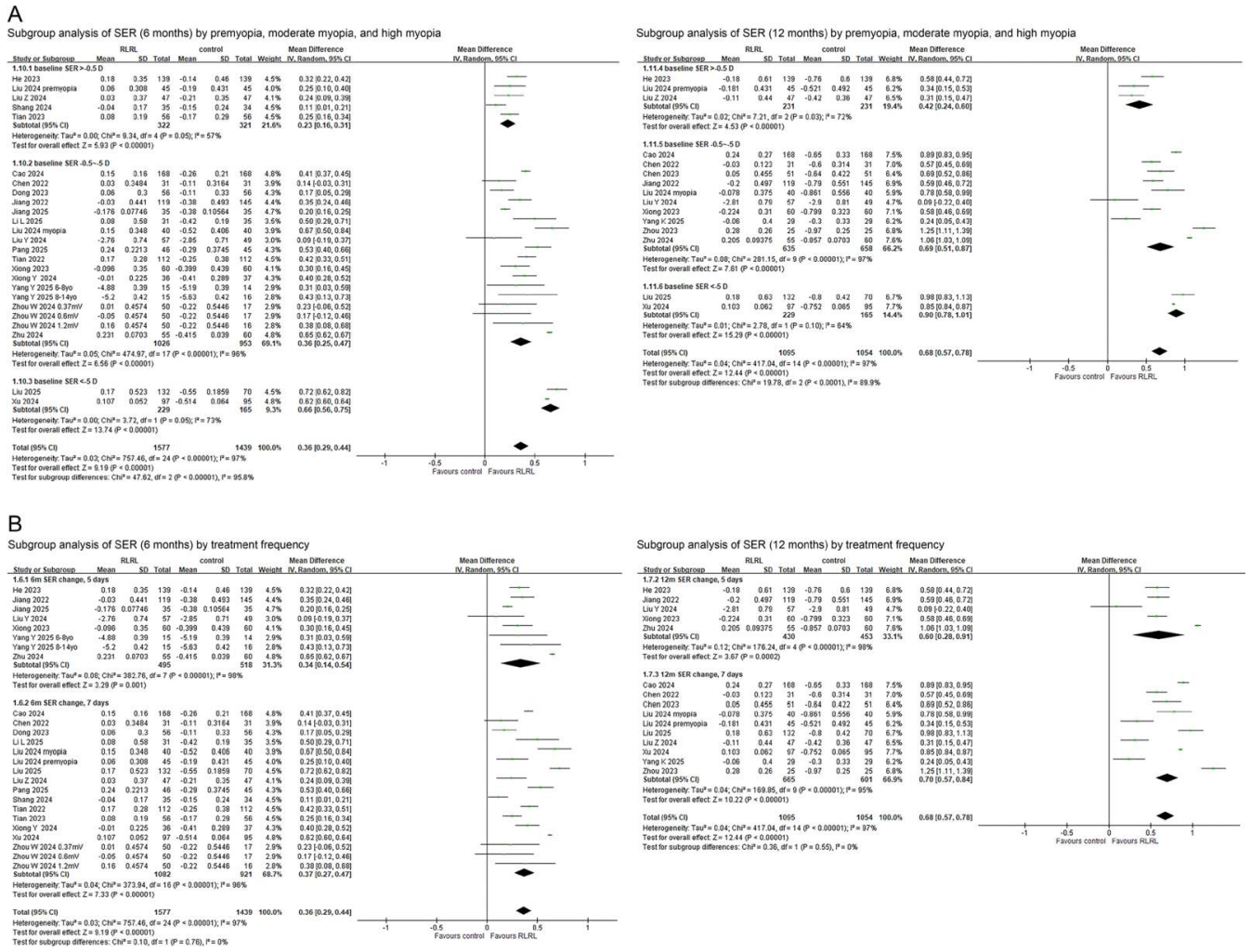


FIGURE 3 | Subgroup analysis of the effect of RLRL therapy on SER. (A) Subgroups based on baseline SER: Premyopia (greater than −0.5 D), moderate myopia (−0.5 to −5.0 D), and high myopia (less than −5.0 D). (B) Subgroups based on treatment frequency: 5 days and 7 days per week.

elongation, and enhanced choroidal thickening. No significant adverse effects were reported or changes on ACD, CCT, or LT were discovered, and the visual acuity outcomes consistently favoured RLRL therapy, supporting its overall efficacy and clinical safety in the studied populations. Based on the GRADE assessment, RLRL therapy can be strongly recommended for improving SER and ChT, while a conditional recommendation is made for AL. Accordingly, this recommendation should be interpreted in the context of stakeholder values, treatment accessibility, and resource considerations, warranting further studies to evaluate the long-term cost-effectiveness and patient-centered outcomes of RLRL therapy.

Although previous meta-analyses of RLRL therapy have provided key insights, their scope differs from that of the present study (Table 5 and Figure S5) [57–74]. Several studies have used network meta-analyses to compare RLRL therapy with other treatment modalities such as atropine eye drops, orthokeratology lenses, and specialised lenses to develop treatment rankings [69–74]. Other studies have emphasised limitations in geographic generalizability [62, 71]. One study extended its focus to premyopic populations, highlighting preventive applications [60], and two studies reported adverse clinical events such as photophobia and corneal infiltrates [63, 71]. Notably, our meta-analysis synthesised the largest RCT-only evidence base to date (28 RCTs involving 3573 participants), demonstrated uniform duration-dependent efficacy, and uniquely incorporated biometric safety outcomes, including corneal, lens, and anterior chamber parameters. Other reviews have emphasised breadth through comparative rankings or prevention in premyopia. By contrast, our analysis provided depth by delivering a definitive efficacy and structural safety assessment based exclusively on high-level RCT evidence. Therefore, these results support the clinical value of RLRL therapy for paediatric myopia but also highlight the importance of long-term monitoring for safety, as well as for assessing the durability of treatment effects and potential post-treatment consequences following cessation.

RCTs investigating RLRL therapy have generally reported no significant changes in anterior segment biometric parameters such as ACD, CCT and LT, and they have also reported no associated complications. Consistent with these reports, our pooled analysis revealed that the CCT and LT remained stable, with no major differences discovered between RLRL therapy and control groups. Although a small reduction in ACD was observed when pooling data from four RCTs ($n = 554$ at 6 months; $n = 778$ at 12 months), the absolute magnitude of this change (approximately 0.01 mm) was minimal and fell within the expected

TABLE 3 | Adverse events reported in the included trials.

Author (year)	Adverse events
Cao K et al. (2024) [13]	Seeing afterimages for a long time ($n=3$) Feeling strong light ($n=4$)
Chen Y et al. (2022) [16]	Dropped out because of the uncomfortable brightness ($n=1$) Dizziness ($n=1$)
Chen H et al. (2023) [32]	No adverse complaints during the study were reported
Deen N 2025 [33]	No severe adverse events were reported
Dong J et al. (2023) [34]	No treatment-related adverse events were reported
He X et al. (2023) [35]	Dropped out because of afterimages lasting > 6 min ($n=2$)
Jiang Y et al. (2022) [30]	Dropped out because of the uncomfortable brightness ($n=2$)
Jiang Z et al. (2025) [14]	No toxic side effects on retinal or optic nerve function
Li L et al. (2025) [37]	No adverse events were reported
Liu G et al. (2024) [39]	Feeling strong light ($n=3$)
Liu G et al. (2025) [38]	No severe adverse events were reported
Liu Y et al. (2024) [40]	Corneal staining ($n=7$) Conjunctival inflammation ($n=6$) Visual disturbances ($n=2$) Photophobia ($n=3$)
Liu Z et al. (2024) [41]	Intolerance to bright light and dry eye symptoms ($n=2$)
Pang X et al. (2025) [17]	Uncomfortable brightness ($n=1$)
Shang L et al. (2024) [42]	No fundus structural change was found during the study
Tian L et al. (2022) [36]	No adverse events were reported
Tian L et al. (2023) [43]	No adverse events were reported
Wei H et al. (2025) [44]	No adverse events or fundus abnormalities were reported
Xiang K et al. (2025) [15]	Nil
Xiong R et al. (2023) [46]	Nil
Xiong R et al. (2024) [45]	Eye pain and tearing ($n=1$) Eye pain, redness, and tearing after lens insertion ($n=3$) Dry eye ($n=1$) Eye itching ($n=2$) Contact lens–induced papillary conjunctivitis ($n=3$) Corneal staining, grade i ($n=7$)
Xiong Y et al. (2024) [47]	No severe adverse events were reported
Xu Y et al. (2024) [48]	Conjunctivitis ($n=1$)
Yang K et al. (2025) [49]	No safety concerns were identified
Yang Y et al. (2025) [50]	Abnormal light spots ($n=3$) Bright spots ($n=6$) Dark spots ($n=3$)
Zhou L et al. (2023) [51]	Reversible vision loss and afterimage ($n=23$)
Zhou W et al. (2024) [52]	No adverse events were reported
Zhu M et al. (2024) [53]	Dropped out because of afterimages lasting > 6 min ($n=1$)

range of physiological fluctuation and instrument repeatability. Evidence suggests that short- to medium-term RCT data have not demonstrated significant changes in intraocular pressure or clinically detectable anterior segment pathology, and no consistent structural retinal abnormalities were reported within the follow-up periods studied [13]. This finding is therefore unlikely to represent a clinically meaningful anatomical alteration. Collectively, the available RCT evidence suggests that, within a follow-up period of up to 1 year, RLRL therapy does not induce clinically significant anterior segment structural instability.

It is important to emphasise that the safety parameters quantitatively synthesised in this meta-analysis were largely confined to anterior segment biometry and trial-reported adverse events. Retinal structural outcomes were insufficiently and inconsistently reported across trials to permit quantitative pooling. Only a small number of studies mentioned retinal thickness assessments, and the heterogeneity of reporting precluded formal meta-analysis [38, 42].

Beyond RCT-reported outcomes, broader retinal safety concerns have been raised in the literature. A recent laboratory-based evaluation assessed the optical output of commercially available RLRL devices in relation to American National Standards Institute (ANSI) ophthalmic instrument standards and reported that certain laser-based systems reached group 1 maximum permissible exposure limits within exposure durations shorter than the recommended 180-s treatment time under specific modelling assumptions, including larger pupil diameters and fixed retinal spot sizes [75]. Notably, such modelling studies are conducted under controlled laboratory conditions and do not directly establish the occurrence of clinically observed retinal injury. In addition, a recent multicentre cohort study of 52 children using high-resolution adaptive optics scanning laser ophthalmoscopy reported an association with reduced cone density in the parafoveal region among children who had undergone RLRL therapy for more than 1 year, with abnormal parafoveal reflectivity signals and a transient OCT-detected cystoid change observed in a small number of cases [76]. Although causality cannot be inferred from observational data and treatment protocols varied, these findings suggest that potential cellular-level retinal effects warrant further investigation. Taken together, these emerging findings indicate that retinal safety remains an evolving and actively debated area of investigation. While

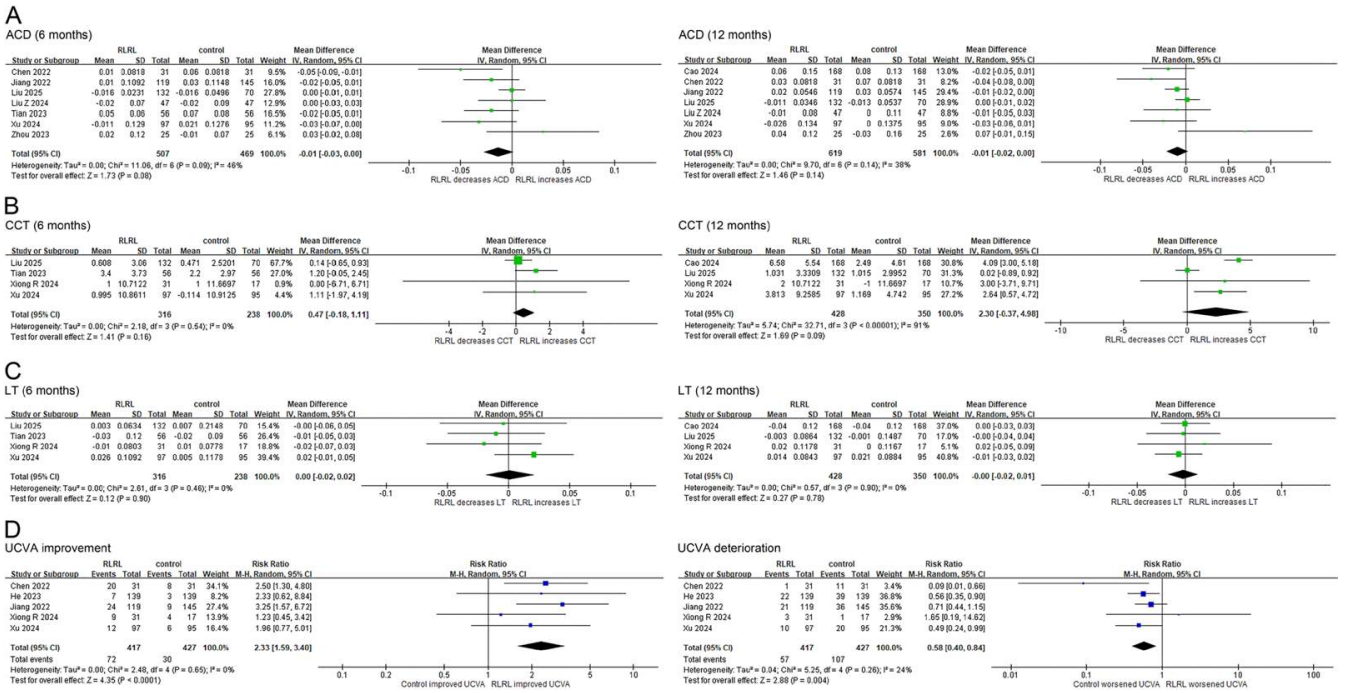


FIGURE 4 | Effects of RLRL therapy on ACD, CCT, LT, and UCVA. (A) Pooled MDs for ACD at 6 and 12 months. (B) Pooled MDs for CCT at 6 and 12 months. (C) Pooled MDs for LT at 6 and 12 months. (D) Relative risks for UCVA improvement (greater than 2-line improvement) and worsening (greater than 2-line worsening) at 12 months.

short-term RCT findings suggest structural and clinical tolerability within 1 year, definitive conclusions regarding cumulative phototoxicity or long-term retinal cellular effects require independently conducted, long-duration studies incorporating standardised retinal imaging and functional assessments.

The heterogeneity across the included trials may partly reflect differences in study populations, intervention parameters, and comparator groups. The patient characteristics differed across the included trials, with ages ranging from preschool to adolescence and baseline refractive errors ranging from premyopia to high myopia. The intervention protocols also differed, particularly in terms of treatment duration (3–12 months), treatment frequency (5 vs. 7 days per week), and light intensity, although our longitudinal and subgroup analyses specifically addressed the impacts of treatment duration, treatment frequency, and baseline refractive error on outcomes. Additionally, variability in the control group contributed to heterogeneity, particularly because some trials compared RLRL therapy with SVSs, whereas others involved low-dose atropine, orthokeratology, or alternative light dosages. These differences in populations, treatment regimens, and comparator interventions may account for the residual variability observed across the study findings despite our subgroup analyses.

This meta-analysis has several key strengths. First, it synthesises the largest RCT-only evidence base to date, incorporating a substantial number of recently published trials and thereby providing an updated and methodologically rigorous assessment. Second, it extends beyond efficacy outcomes by integrating both quantitative analyses of ocular structural parameters and systematic trial-level safety reporting, offering a more comprehensive appraisal of short- to medium-term safety. Third, it characterises duration-dependent treatment effects using both stratified analyses and meta-regression, rather than relying solely on single pooled estimates. Finally, by applying the GRADE framework, this study provides outcome-specific, certainty-based recommendations to support clinically informed interpretation.

This study has some limitations. First, almost all of the included trials were conducted in China, which raises concerns regarding geographic generalizability. One included RCT, however, was performed in a multiethnic cohort comprising Indian, Caucasian, African, and Middle Eastern participants and demonstrated efficacy outcomes comparable to those observed in Chinese populations [33]. Second, unlike network meta-analyses, this review did not provide direct comparisons of RLRL therapy with other established modalities such as atropine or orthokeratology. All included trials used the same baseline intervention in both groups, allowing the isolated effect of RLRL to be evaluated rather than a direct comparison with existing modalities. Moreover, the present study was limited to outcomes reported within a follow-up period of up to 1 year, reflecting the current lack of longer-term RCT data. Consequently, evidence regarding long-term outcomes and post-cessation effects remains limited. The durability of RLRL's benefits after treatment discontinuation, the potential for refractive regression, and the occurrence of late-onset structural changes have yet to be fully characterised. Finally, the safety evaluation in this meta-analysis was restricted to parameters systematically reported within RCT settings, primarily anterior segment biometry and trial-reported clinical tolerability. Comprehensive retinal safety domains—including detailed structural imaging, functional retinal assessments, and photobiological exposure analyses—were not consistently

TABLE 4 | Summary of findings and certainty of evidence.

Summary of findings

Low-Level Red-Light Therapy compared to control for Slowing Myopia Progression in Children

Patient or population: Slowing Myopia Progression in Children
Setting:
Intervention: Low-Level Red-Light Therapy
Comparison: control

Outcomes	Anticipated absolute effects[d] (95% CI) Risk with control risk with low-level red-light therapy	Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
Spherical Equivalent Refraction of 12 months (SER) assessed with: D	MD 0.68 D higher (0.57 higher to 0.78 higher)	—	2149 (14 RCTs)	⊕⊕⊕○ Moderate[a]	Low-Level Red-Light Therapy is likely to reduce spherical equivalent refraction deterioration over 12 months.
Axial length of 12 months (AL) assessed with: mm	MD 0.3 mm lower (0.37 lower to 0.23 lower)	—	2197 (15 RCTs)	⊕⊕○○ Low[b,c]	Low-Level Red-Light Therapy may reduce axial elongation over 12 months.
Subfoveal choroidal thickness (SFChT) assessed with: μm	MD 26.7 μm higher (18.64 higher to 34.77 higher)	—	2002 (13 RCTs)	⊕⊕⊕○ Moderate[c]	Low-Level Red-Light Therapy is likely to reduce subfoveal choroidal thickness deterioration over 12 months.

Note: GRADE Working Group grades of evidence. High certainty: we are very confident that the true effect lies close to that of the estimate of the effect. Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect. Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.
Abbreviations: CI, confidence interval; MD, mean difference.
[a]The included studies were similar in design and population, and the direction of effect was consistent with clinical relevance but $I^2 > 90\%$.
[b]Egger's test p value < 0.05.
[c]The direction of effect was inconsistent with clinical relevance and $I^2 > 90\%$.
[d]The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).

TABLE 5 | Characteristics of published systematic reviews and meta-analyses.

Author (year)	Population	Intervention	Comparison	Outcome	Inclusion or exclusion	Study types	Number of studies	Intervention (*n*)	Control (*n*)	SER change (MD [95% CI])	AL change (MD [95% CI])	ChT change (MD [95% CI])	Adverse events	Recommend? (O/△/X)
Head to head														
Amaral D et al. (2024) [57]	3–15 years with myopia	RLRL	Control	AL, SER	Excluded: Studies without a control group	RCT: 4 Cohort: 1	5	357	328	Mixed: 0.58 [0.33, 0.83]	Mixed: −0.33 [−0.52, −0.13]			O
Deng B et al. (2024) [58]	7–15 years without a history of ocular or systemic diseases, ocular surgical trauma, amblyopia, or strabismus	RLRL	SVS, atropine, or sham device	AL, SER Adverse events	Included: Only English or Chinese articles Excluded: Study subjects other than humans Studies lacking full text or complete data	RCT	6	393	400	Mixed: 0.46 [0.32, 0.6]	Mixed: −0.22 [−0.28, −0.16]			O
Fan H et al. (2025) [59]	Children with myopia or pre-myopia	RLRL	Control	AL, SER, SFChT	Excluded: Studies enrolling adult populations Studies applied RLRL in combination with other myopia control interventions	RCT: 15 Cohort: 5	20	1405	1233	6 months: 0.40 [0.31, 0.50] 12 months: 0.61 [0.47, 0.76] 24 months: 1.33 [0.62, 2.03]	6 months: −0.22 [−0.25, −0.18] 12 months: −0.30 [−0.36, −0.24] 24 months: −0.61 [−0.71, −0.52]	6 months: 31.21 [22.03, 40.38] 12 months: 29.72 [19.53, 39.92]		O
Leber M et al. (2025) [60]	Children with premyopia (SER < 0.75 D and SER > −0.50 D)	Intense foveal red light	Control	Myopia incidence, AL, SER, and SFChT	Included: Only English articles Excluded: Studies without a control group or with a potentially overlapping population	RCT	4	326 eyes	314 eyes	6 months: 0.27 [0.23, 0.32] 12 months: 0.36 [0.27, 0.46]	6 months: −0.12 [−0.16, −0.09] 12 months: −0.18 [−0.23, −0.14]	Mixed: 22.34 [5.45, 39.24]		O
Liu L et al. (2025) [61]	< 18 years with myopia	RLRL	Conventional myopia control treatment	AL, SER, or visual acuity	Included: Only English or Chinese articles Excluded: Studies providing follow-up data spanning < 3 months Studies lacking complete data Studies comparing RLRL therapy with other nontraditional methods Research articles with inaccessible full texts	RCT: 4 Cohort: 2 Nonrandomized clinical trial: 1	7	349	342		Mixed: −0.16 [−0.21, −0.12]	Mixed: 30.79 [17.28, 44.30]		O

(Continues)

TABLE 5 | (Continued)

Author (year)	Population	Intervention	Comparison	Outcome	Inclusion or exclusion	Study types	Number of studies	Intervention (*n*)	Control (*n*)	SER change (MD [95% CI])	AL change (MD [95% CI])	ChT change (MD [95% CI])	Adverse events	Recommend? (O/△/X)
Sobol M et al. (2025) [62]	6–16 years with myopia (cycloplegic SER ≤ −0.5 D); without ocular or systemic diseases Astigmatism ≤ 1.5 D, anisometropia ≤ 1.5 D, IOP 10 to 21 mmHg	RLRL	SVS alone	AL and SER	Included: Only human studies, English articles Excluded: Participants who received any other myopia therapy apart from wearing standard corrective spectacles Studies reporting results only as median and range or IQR; mean value but without SD or CI	RCT: 9 Retrospective case series: 1	10	839	785	6 months: 0.18 [−0.03, 0.39] 12 months: 0.05 [−0.31, 0.42]	6 months: 0 [−0.1, 0.1] 12 months: −0.01 [−0.16, 0.13]			O
Tang J et al. (2023) [63]	< 18 years with myopia	Red light used	Control	AL, SER, SFChT, and UCVA	Excluded: Animal experiments and studies with low quality	RCT: 5 Cohort: 2 Nonrandomized clinical trial: 1	8	478	456	6 months: 0.54 [0.33, 0.74] 12 months: 0.61 [0.52, 0.7]	6 months: −0.29 [−0.42, −0.16] 12 months: −0.29 [−0.37, −0.21]	6 months: 31.66 [7.48, 55.84] 12 months: 26.68 [15.69, 37.67]	Perceiving the light as too bright	O
Ullah S et al. (2025) [64]	3–18 years with myopia (cycloplegic SER ≤ −0.5 D)	SVS + RLRL	SVS alone	AL and SER	Included: Only English articles Excluded: Animal studies, in vitro studies, and quasiexperimental studies Studies with follow-up durations of < 12 months	RCT	10	824	890	Bayesian pooled effect size 12 months: 1.52 [0.10, 2.74]	Bayesian pooled effect size 12 months: −0.95 [−1.50, −0.35]			O
Wang F et al. (2023) [65]	Children with myopia	RLRL	Control (without RLRL)	AL and SER	Included: Only English or Chinese articles Excluded: Studies lacking full text	Parallel-group RCT	7	474	557	6 months: 0.43 [0.16, 0.69] 12 months: 0.67 [0.48, 0.86]	6 months: −0.19 [−0.28, −0.10] 12 months: −0.32 [−0.46, −0.18]			O
Yang X et al. (2025) [66]	< 18 years	RLRL	Sham device, SVS, or continued usual activities	AL, SER, and SFChT	Included: Only English articles Excluded: Studies with follow-up durations of < 6 months	RCT	8	788	690	Mixed: 0.49 [0.27, 0.71]	Mixed: −0.24 [−0.32, −0.17]	Mixed: 28.16 [18.78, 37.55]		O
Youssef M et al. (2024) [67]	3–18 years with myopia (cycloplegic SER ≤ −0.5 D)	SVS + RLRL	SVS alone	AL, SER, and SFChT	Included: Only English articles Excluded: Animal studies and in vitro studies Studies with follow-up durations of < 3 months	RCT	5	407	426	6 months: 0.46 [0.26, 0.65] 12 months: 0.63 [0.52, 0.73]	6 months: −0.21 [−0.28, −0.15] 12 months: −0.31 [−0.42, −0.19]	6 months: 34.75 [15.26, 54.25]		O

(Continues)

TABLE 5 | (Continued)

Author (year)	Population	Intervention	Comparison	Outcome	Inclusion or exclusion	Study types	Number of studies	Intervention (*n*)	Control (*n*)	SER change (MD [95% CI])	AL change (MD [95% CI])	ChT change (MD [95% CI])	Adverse events	Recommend? (O/△/X)
Zhang H et al. (2023) [68]	6–18 years with myopia (cycloplegic SER ≤ −0.5 D); without a history of ocular surgery, trauma, or similar conditions	SVS + RLRL	SVS alone	AL and SER	Included: Only English or Chinese articles Excluded: Studies involving other previously used myopia control methods Studies with low quality, with incomplete data, or with no access to the original data	RCT	7	486	552	12 months: 0.68 [0.52, 0.84]	12 months: −0.40 [−0.56, −0.23]			O
Network meta-analysis														
Lawrenson J et al. (2025) [69]	< 18 years, SER ≤ −0.5 D	Means to achieve myopia control	Placebo, SVS, or contact lenses	AL and SER		RCT	81			12 months: 0.84 [0.63, 1.05]	12 months: −0.35 [−0.41, −0.29]			O
Lee S et al. (2025) [70]	Children without myopia (SER ≥ −0.50 D) and without congenital or severe ophthalmological conditions or a history of eye surgery	Means to achieve myopia control	Placebo	Myopia incidence, AL, and SER		RCT	19			12 months: 0.48 [0.38, 0.59]	12 months: −0.23 [−0.27, −0.19]			O
Schmidt D et al. (2025) [71]	6–18 years with myopia (cycloplegic SER ≤ −0.5 D in at least one eye); included studies in which at least 90% of the population was within the defined age range of eligibility	Means to achieve myopia control	Any other intervention or observation/ placebo	AL; adverse events	Included: Only peer-reviewed full-text studies, English articles Excluded: Studies with an intervention period of < 1 year (10–14 months) Studies with no data on a comparison group	RCT	74				12 months: −0.33 [−0.4, −0.25]		Photochemical and thermal retinal damage, changes in photoreceptor outer segments, and decrease in visual acuity	O
Zaabaar E et al. (2024) [72]	< 18 years with myopia (SER ≤ −0.5 D) and without ocular comorbidities such as strabismus and amblyopia	Nonsurgical myopia control interventions	SVS, contact lenses, placebo treatment, or a combination of all modalities	AL and SER	Included: Only peer-reviewed English articles Excluded: Studies with low quality Studies with no quantitative data Correlational studies and investigations related to surgical interventions	RCTs and nonrandomized comparative studies	38			6 months: 0.47 [0.31, 0.63] 12 months: 0.77 [0.50, 1.04]	6 months: −0.27 [−0.37, −0.17] 12 months: −0.44 [−0.31, −0.28]			O

(Continues)

TABLE 5 | (Continued)

Author (year)	Population	Intervention	Comparison	Outcome	Inclusion or exclusion	Study types	Number of studies	Intervention (*n*)	Control (*n*)	SER change (MD [95% CI])	AL change (MD [95% CI])	ChT change (MD [95% CI])	Adverse events	Recommend? (O/△/X)
Zhang G et al. (2023) [73]	6–18 years with myopia	Myopia control interventions	Control group	AL and SER	Included: Only English or Chinese articles Excluded: Animal studies and in vitro studies Studies with follow-up durations of < 1 year	RCT	80			Mixed: 0.59 [0.06, 1.1]	Mixed: −0.25 [−0.86, 0.39]			△ (AL nonsignificant)
Zheng Z et al. (2025) [74]	3–16 years with myopia	0.01% atropine, orthokeratology, RLRL, or their combination	SVS, placebo, or no treatment	AL and SER	Included: Only English articles Excluded: Studies with treatment durations of < 6 months Studies without AL or SER as the outcome or studies not providing data suitable for meta-analysis	RCT	41			12 months: 0.76 [0.54, 0.98]	12 months: −0.31 [−0.39, −0.24]			O

Note: Symbols in 'Recommend?' column: O: recommended; △: conditional/uncertain recommendation; X: not recommended.

Abbreviations: ACD, anterior chamber depth; AL, axial length; ChT, choroidal thickness; CI, confidence interval; IOP, intraocular pressure; IQR, interquartile range; MD, mean difference; RCT, randomized controlled trial; RLRL, repeated low-level red-light; SD, standard deviation; SER, spherical equivalent refraction; SFChT, subfoveal choroidal thickness; SVS, single-vision spectacles; UCVA, uncorrected visual acuity.

or sufficiently reported across the included trials to permit quantitative synthesis. Future large-scale multicentre trials incorporating standardised retinal imaging, functional testing, and extended follow-up are required to enable a more complete evaluation of sustained efficacy, long-term ocular safety, and real-world applicability.

In conclusion, this meta-analysis suggests that RLRL therapy is associated with consistent, duration-dependent benefits in slowing myopia progression and axial elongation, and in enhancing choroidal thickness in children over a treatment period of up to 1 year. Short- to medium-term structural parameters and clinical tolerability appear acceptable within the follow-up durations studied; however, long-term retinal safety and cumulative photobiological effects warrant careful consideration in clinical practice. Based on GRADE assessment, moderate-certainty evidence supports the use of RLRL therapy for improving SER and ChT, whereas evidence for AL remains of lower certainty. Further multicentre studies with extended follow-up and comprehensive retinal safety assessments are required to clarify long-term efficacy, cumulative photobiological safety, and generalisability across diverse populations.

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Ethics Statement

This study did not meet the criteria for human subjects research as defined by our institution as it did not include patient data. Therefore, it did not require institutional review board approval or informed consent. Our study adhered to the Declaration of Helsinki.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Pooled MDs of ocular parameters with RLRL therapy at different follow-up time points: (A) SER, (B) AL, and (C) ChT. **Figure S2:** Subgroup analysis of the effects of RLRL therapy on AL and ChT, stratified by baseline SER with premyopia (greater than −0.5 D), moderate myopia (−0.5 to −5.0 d), and high myopia (Less Than −5.0 D). **Figure S3:** Subgroup analysis of the effects of RLRL therapy on AL and ChT, stratified by treatment frequency with 5 versus 7 days per week. **Figure S4:** Funnel plots of 12-month SER, AL, and ChT. **Figure S5:** PRISMA flow diagram of the literature search process for published systematic reviews and meta-analyses. **Table S1:** PRISMA checklist. **Table S2:** Search strategies. **Table S3:** Sensitivity analysis of SER, AL, and ChT at 12 months, performed using the leave-one-out method.